Clustering Asthma Patients: Unveiling Hidden Patterns Using PEFR and Demographic

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Problem Statement

We received data on 114 patients with severe asthma, a chronic respiratory condition. When observing the patient data, we noticed there may be correlations between higher PEFR values and other provided demographic information (e.g. age, weight, height, smoking status, occupation, living environment). Our project aims to clearly identify and determine these correlations. To do so, we will calculate summary statistics of each patient data and combine them with demographic information to find trends among the data. Through these trends, our project will ultimately cluster patients into different categories. Ultimately, we will develop a prediction model which will predict the patient's group depending on demographic information.

Data Sources

Division of Allergy and Respiratory Medicine at Soonchunhyang University Bucheon Hospital, South Korea.

PEFR_asthma_114_medinfo_07.15 Dataset: 14 columns, 1 rows

- Demographic Attributes: age, sex, smoking status, height, weight, BMI, occupation
- Additional patient profile information (UID, BCODE)

99 Individual Patient Data Files: 14 columns, varying rows

PEFR Readings (AM, PM)

Objectives

1. Data Cleaning & Preprocessing: Handle missing data through appropriate imputation techniques and standardize demographic data, ensuring accuracy.
2. Summary Statistics: Find summary statistics (min, max, mean, median, etc.) for the longest chunk of data (without missing values) for each patient. Add these statistics to the table with demographic information (age, height, weight, etc.).
3. Clustering Analysis: Group patients though correlation and feature analysis. The quality of the clusters will then be validated using well-known metrics.

Challenges

- Handling Missing Data
    - Determine appropriate imputation strategies to ensure imputed data does not distort real trends
    - Address missing PEFR and demographic data by imputation or dropping data depending on the amount of missing data (gaps) and what kind of missing data (completely random, random, not random).
- Correlation Analysis and Feature Selection: finding correlations and trends to identify which demographic or environmental factors are relevant for predicting PEFR statistics
- Group Prediction Model: ensuring features are chosen properly to develop clusters and develop an accurate prediction model
- Outliers: Differentiating between true variability in PEFR and erroneous data entries.

Data Preprocessing and Cleaning Plan

Translation: translating occupation and address to English

Handling Missing Data

1. Determining if there are substitutes (0, -, 'unknown') for missing data and changing them to be NaN
2. Determine the size of the missing data gaps
    a. For gaps less than 14 days, impute data using:
        - Linear Regression: Predict missing values based on trends.
        - Forward/Backward Fill: Use adjacent PEFR data points.
        - Nearest Fill: Impute based on similar patients' PEFR trends.
    b. Find the longest piece of consecutive data
3. Impute missing data as appropriate
    a. Try different methods: linear regression, forward fill, backward fill, nearest fill
    b. Visualize imputed data and original data to see which method is best

Handling Outliers

1. Determine erroneous data
2. Determine possible outliers
    a. Handling extreme contrasts in PEFR values for one patient (e.g. 300 – 800)

b. Statistical Techniques: Use interquartile range (IQR) and z-scores to detect outliers in PEFR and demographic data.

Correlation and Feature Selection

- Pearson/Spearman Correlation: Analyze the relationships between PEFR and factors such as age, BMI, smoking, etc.
- Feature Importance: Use linear regression and random forest models to identify key predictive features.

Training and Testing

- Model Selection: Compare multiple models, including logistic regression, random forest, and decision trees.
- Evaluation Metrics: Measure model performance using accuracy, precision, recall, r-squared.
- Preventing Overfitting: Use cross-validation and regularization techniques.
- Evaluation Metrics: Measure model performance using accuracy, R-squared, and mean absolute error (MAE).